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PURPOSE

Remote wildfire areas often lack real-time monitoring.

SOLUTION

Drone nodes detect fire/smoke, send GPS telemetry, and help estimate fire location on a live ground-station map.

OBJECTIVES

- Detect fire/smoke from aerial video
- Transmit drone GPS, heading, and alert status
- Locate fire using 2+ drone observations
- Display alerts on a live map

TARGET USERS

- Wildfire operations coordinators
- Fire-response and forestry field crews
- Environmental researchers
- Future engineering teams

SOFTWARE

Ground Station Interface

- Displays drone positions, alerts, and estimated fire markers

Vision Pipeline

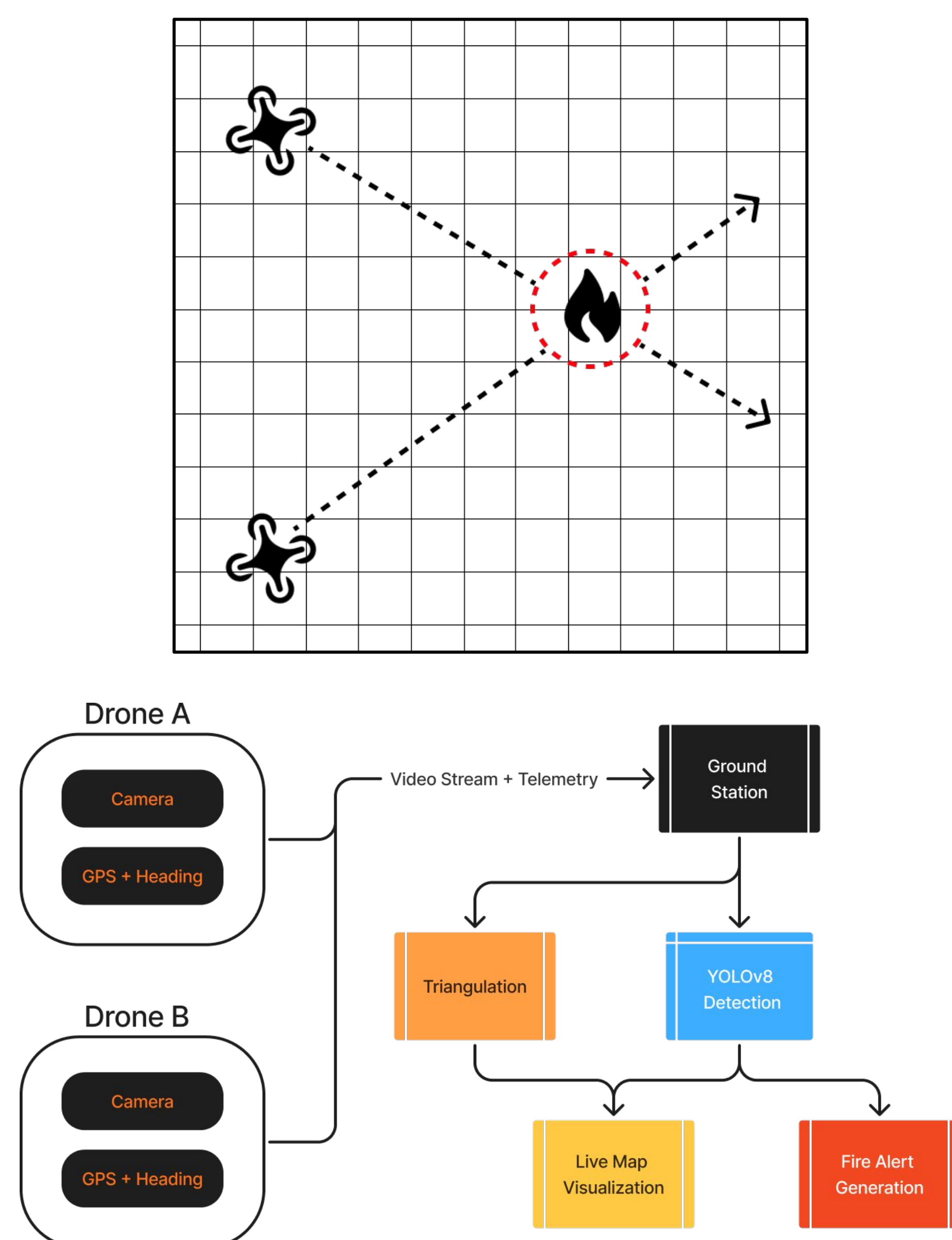
- YOLOv8 fire/smoke detection

Localization

- Estimates GPS coordinates of detected fire events

Telemetry Backend

- Parses real-time ESP32 telemetry messages
- Updates active drone and fire-event state



HARDWARE

Drone Node

- ESP32 flight/telemetry microcontroller
- ESP32-CAM video module
- GPS receiver for position data

Wireless Communication

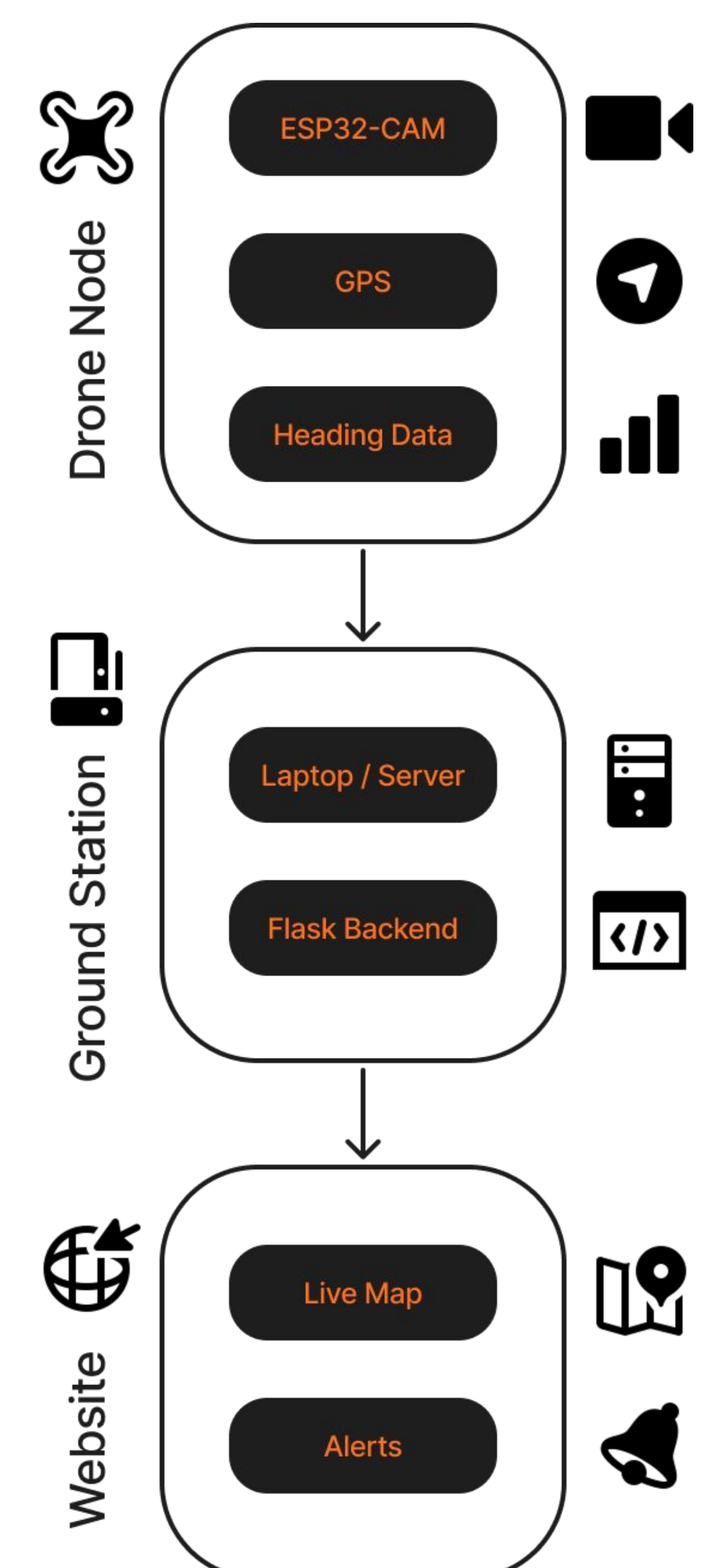
- ESP-NOW broadcast telemetry between drone nodes
- Wi-Fi connection to ground station

Ground Station

- Receives telemetry and fire alerts
- Displays live map and detection results

Sensor Data Collected

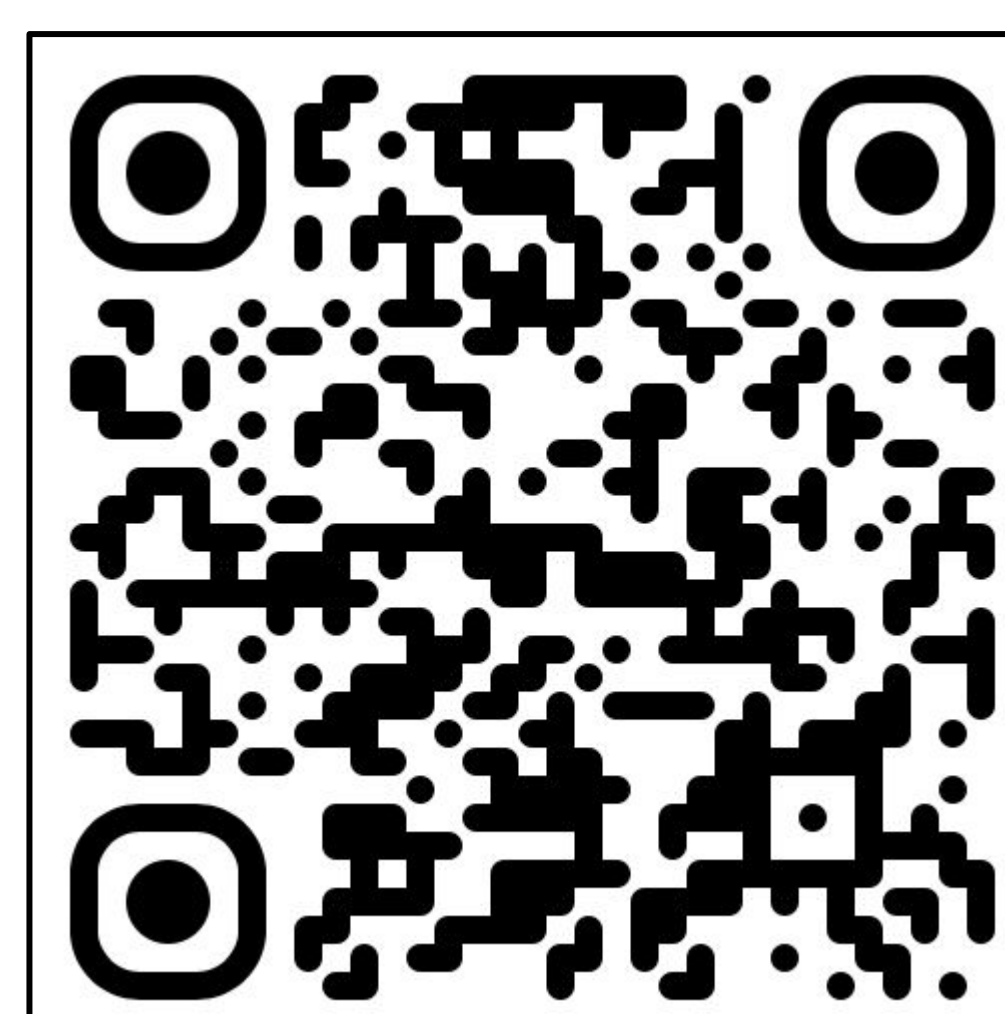
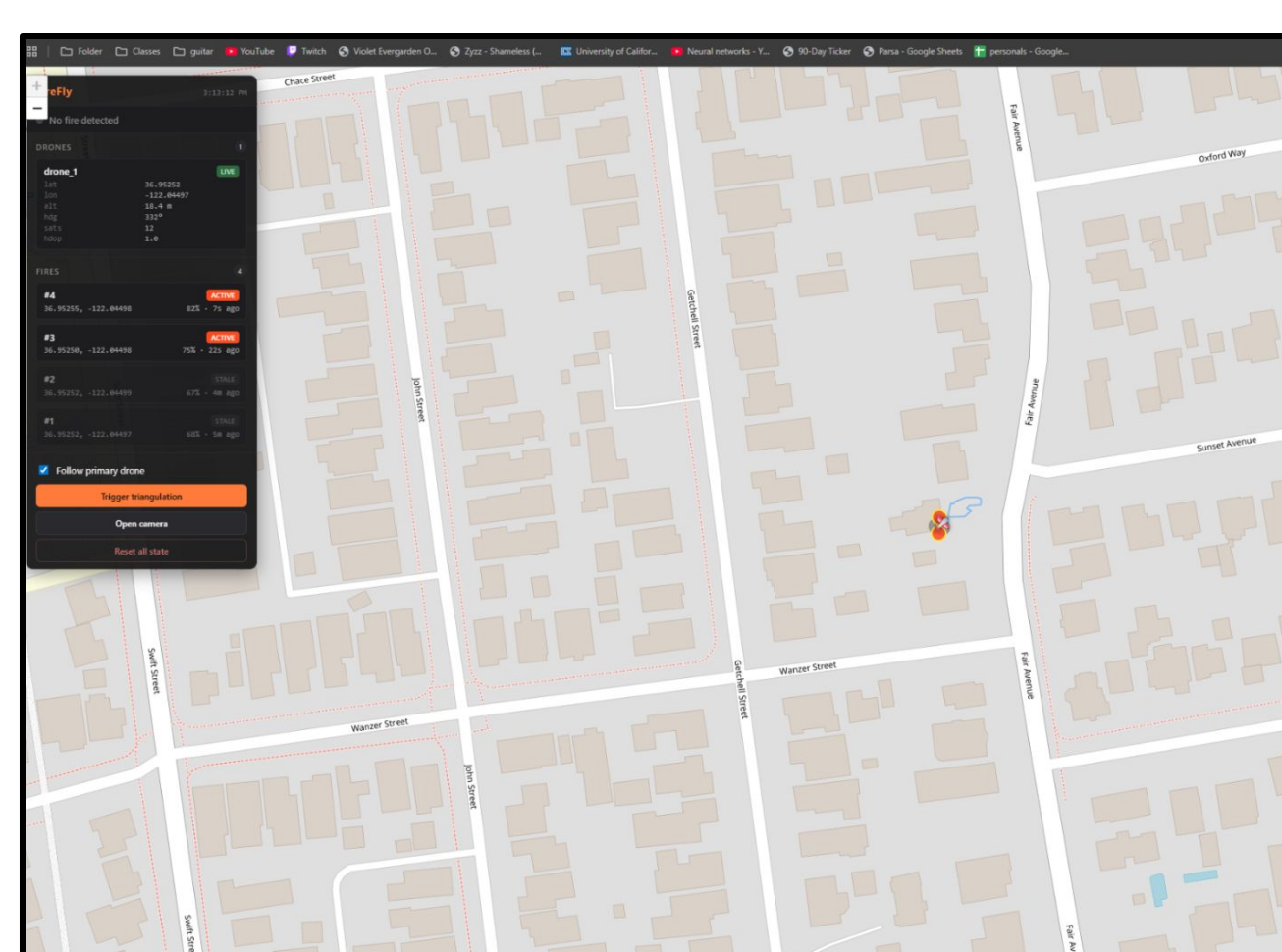
- Camera stream for fire/smoke detection
- GPS coordinates



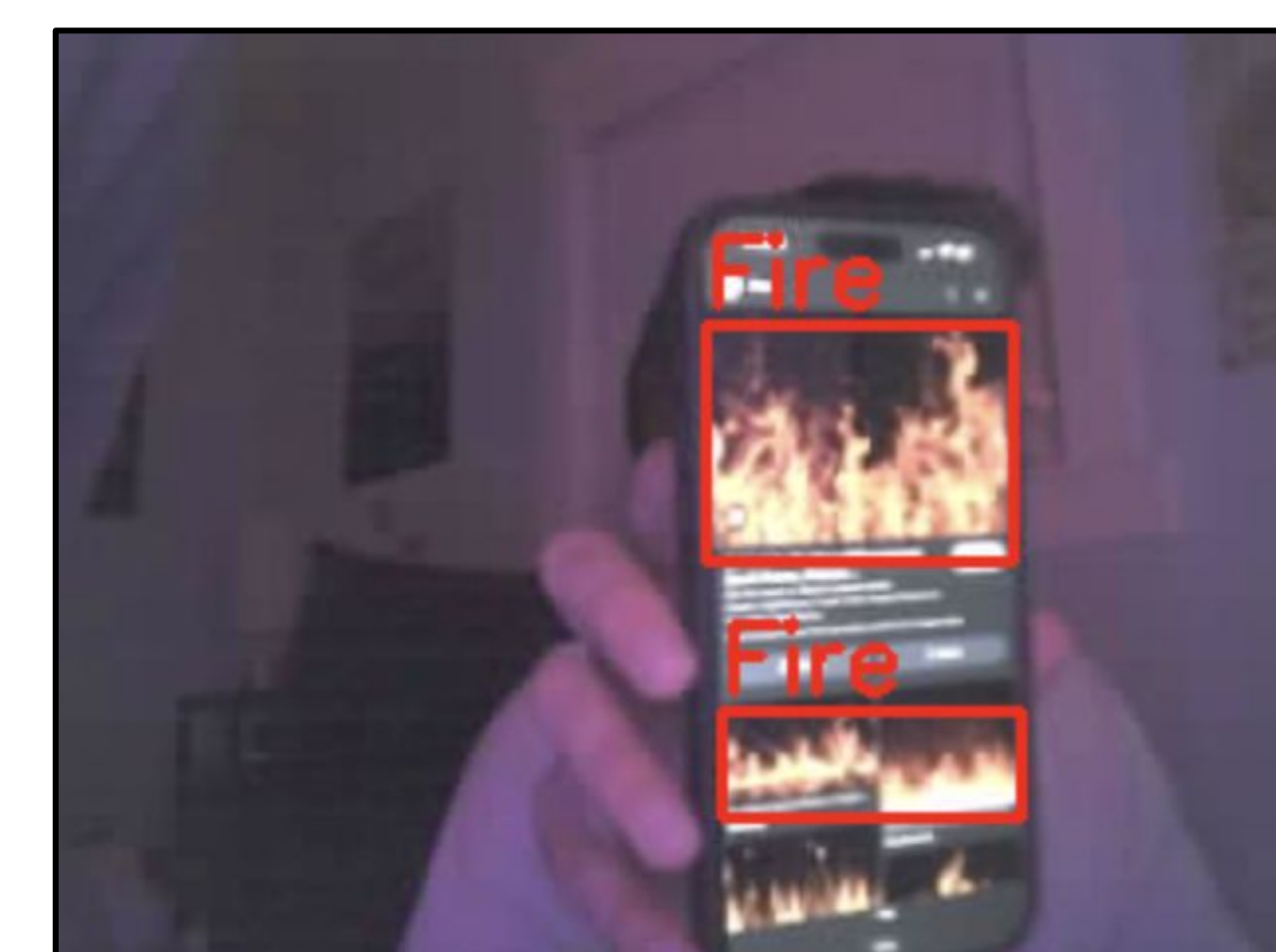
DEMO AND VALIDATION

Ground Station Map

- Drone telemetry and estimated fire coordinates are displayed in real time



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Fire Detection View

- YOLOv8 identifies sustained fire and smoke from the camera stream